PARTICIPANT CHALLENGE BRIEF

Ghost Data Still Exists

HackerRank Orchestrate | October 2026

THE CHALLENGE

Build an AI-powered deletion-audit agent that determines whether a person's data was actually deleted, identifies records and processes that still expose or can recreate that data, and produces an ordered, evidence-backed plan to finish or verify deletion.

The scenario

Northstar Shop, an imaginary company, loses $2.4 million after a customer who was supposedly deleted later cannot create a new account because the email is still registered. The investigation finds personal data scattered across secondary databases, support attachments, search indexes, analytics exports, logs, archives, and backups. A synchronization job has even recreated data after the primary account service deleted it.

Your system must look beyond a dashboard's "Complete" label. It must reconstruct what happened across the supplied evidence and determine the true state at the audit time.

What you need to build

- Resolve the person's current and historical identifiers.
- Inspect the supplied records and determine whether deletion was requested and whether it is complete.
- Find remaining copies, exposed personal information, active data flows, and restore risks.
- Apply the deletion policy to decide what must be deleted, redacted, anonymized, or retained.
- Return the ordered remediation steps or the checks that prove deletion is already complete.

When is deletion complete?

A case is COMPLETE only when all four conditions hold:

- Every relevant copy has received the treatment required by deletion_rules.csv.
- No active sync, import, or restore path can recreate the data.
- Retained records contain only the identifiers permitted by the rules.
- Records that must be retained remain available.

IMPORTANT

The workflow's reported status and operation results are claims, not proof. Determine the resource state at audit_at from the available evidence.

01 / INPUTS & OUTPUT CONTRACT

Participant dataset

All participant-facing files are supplied inside the repository's dataset/ directory. The evidence manifest identifies the evidence files; evidence itself may be stored in multiple formats.

File	Purpose
deletion_requests.csv	Cases that must be audited.
sample_deletion_requests.csv	Sample cases.
sample_output.csv	Correct answers for the sample cases; use as a format reference.
user_identities.csv	Current user identities.
user_audit_log.csv	Historical identity changes.
system_activity.csv	Deletion, sync, export, backup, restore, indexing, and related activity.
deletion_rules.csv	Required treatment for each data category.
evidence_files.csv	Manifest of the supplied evidence files.
output.csv	Blank submission template.

Evidence may be PDFs, images, logs, JSON, or SQLite-compatible SQL snapshot exports under dataset/evidence/. SQL captures may include primary, regional, reporting, or failover replicas.

Input format

Each row in deletion_requests.csv is one case with these fields:

```
case_id,email,requested_at,reported_status,reported_at,audit_at,request_text
```

reported_status may be COMPLETE, PARTIALLY_COMPLETE, or NOT_STARTED. It describes the existing workflow's claim; it is not proof of the actual deletion state.

Required output

Create output.csv with exactly these columns, in this order:

```
case_id,deletion_status,steps_json
```

Column	Requirement
case_id	The case being audited.
deletion_status	Exactly one of COMPLETE, INCOMPLETE, or NOT_APPLICABLE.
steps_json	An ordered JSON array of findings, remediation actions, and verification checks.

EXACT ROW COUNT AND ORDER

The final output.csv must contain exactly 518 data rows, sorted by case_id, plus the header row. Produce exactly one row for every request in deletion_requests.csv.

Step object format

```
{
  "step_no": 1,
  "phase": "FINDING",
  "system_id": "account_service",
  "resource_id": "account_005",
  "operation": "REPORT_REMAINING_RECORD",
  "state": "PRESENT",
  "evidence_id": "evidence_01"
}
```

02 / ALLOWED OPERATIONS

Operation reference

Use only the phase, operation, and state combinations listed below. Operation names and state values must match exactly.

Phase	Operation	State	Meaning
FINDING	REPORT_REMAINING_RECORD	PRESENT	A record that should have been deleted still exists.
FINDING	REPORT_EXPOSED_PII	PRESENT	A record may remain, but still contains identifiers that should have been redacted or anonymized.
FINDING	REPORT_ACTIVE_DATA_FLOW	ACTIVE	An active process can copy or recreate the user's data.
FINDING	REPORT_RESTORE_RISK	ACTIVE	A backup or restore path can bring deleted data back.
REMEDIATION	DISABLE_DATA_FLOW	DISABLED	Stop the process that can copy or recreate the data.
REMEDIATION	BLOCK_RESTORE	BLOCKED	Prevent the affected backup from restoring the user's data.
REMEDIATION	DELETE_RECORD	ABSENT	Remove the entire record.
REMEDIATION	REDACT_RECORD	REDACTED	Keep the record but remove the identifying fields required by policy.
REMEDIATION	ANONYMIZE_RECORD	ANONYMIZED	Keep the record after irreversibly unlinking it from the user.
VERIFICATION	VERIFY_RECORD_ABSENT	ABSENT	Confirm that a record no longer exists.
VERIFICATION	VERIFY_RECORD_PRESENT	PRESENT	Confirm that a policy-required retained record still exists.
VERIFICATION	VERIFY_RECORD_REDACTED	REDACTED	Confirm that required identifying fields have been removed.
VERIFICATION	VERIFY_RECORD_ANONYMIZED	ANONYMIZED	Confirm that the record can no longer be linked to the user.
VERIFICATION	VERIFY_DATA_FLOW_DISABLED	DISABLED	Confirm that the relevant data flow is inactive.
VERIFICATION	VERIFY_RESTORE_BLOCKED	BLOCKED	Confirm that the affected data cannot be restored.

DO NOT INVENT OPERATION LABELS

The grader expects the exact allowed values. Keep operation and state pairs consistent with this reference.

03 / AUDIT & ORDERING RULES

Rules for every case

- Resolve the user's current and historical identifiers before searching for records.
- Determine each resource's state at audit_at using the supplied evidence. Reported workflow statuses and operation results are claims.
- Apply deletion_rules.csv to every relevant record, including account tombstones, active data flows, and restore risks.
- Inspect secondary systems and replicas as well as the primary account service.
- Preserve records that policy requires the organization to retain.

Special handling: account tombstones

Some systems preserve an account tombstone so permitted records can continue to reference the same internal user_id. In these records, the stable user reference may remain, but direct identifiers such as name, email, phone, and email hash must be removed.

Required step ordering

Order every steps_json array by phase, then by the operation sequence below. Within each operation group, sort by system_id and then resource_id in ascending order.

Sequence	Phase	Operation order
1	FINDING	REPORT_REMAINING_RECORD → REPORT_EXPOSED_PII → REPORT_ACTIVE_DATA_FLOW → REPORT_RESTORE_RISK
2	REMEDIATION	DISABLE_DATA_FLOW → BLOCK_RESTORE → DELETE_RECORD → REDACT_RECORD → ANONYMIZE_RECORD
3	VERIFICATION	VERIFY_RECORD_ABSENT → VERIFY_RECORD_PRESENT → VERIFY_RECORD_REDACTED → VERIFY_RECORD_ANONYMIZED → VERIFY_DATA_FLOW_DISABLED → VERIFY_RESTORE_BLOCKED

Status-specific requirements

Status	Required steps_json content
COMPLETE	Verification steps only. They must prove the current state.
INCOMPLETE	Findings, the complete remediation plan, and the checks that must pass afterward.
NOT_APPLICABLE	An empty JSON array ([]) because no deletion or deletion verification is authorized.

```
case_000,NOT_APPLICABLE,[]
```

Step numbers start at 1 and increase consecutively. The array order is significant; JSON object-key order and whitespace are ignored by the grader.

Evidence IDs

Findings and verification steps for COMPLETE cases must cite a file_id, activity_id, or audit_id. Proposed remediation and future verification steps for INCOMPLETE cases must use an empty evidence_id string ("").

EVIDENCE DISCIPLINE

Do not treat an old workflow status, a claimed operation result, or an unverified system message as proof. Tie findings and completed-state verification to the supplied evidence.

04 / EVALUATION & DELIVERABLES

How your solution is evaluated

output.csv is compared with hidden canonical answers. Scoring considers:

- Correct deletion_status for each case.
- Resolution of current and historical identities.
- Discovery of remaining records and active recreation or restore risks.
- Correct application of deletion policy and evidence references.
- Completeness and ordering of findings, remediation, and verification steps.

Required submission items

Item	What to submit
code.zip	Runnable solution, prompts, configuration, README, and evaluation/ folder. Do not include the dataset or nested archives.
output.csv	One answer row for every request in deletion_requests.csv; 518 data rows sorted by case_id.
chat_transcript	The development/use transcript generated by your AI coding tool, uploaded as the chat_transcript submission.
AI interview	Explain your approach, architecture, and evaluation.

Evaluation report

Include evaluation/evaluation_report.md in code.zip. It must document:

- Case-level status accuracy.
- Step-level and field-level accuracy.
- Sequence accuracy.
- At least three additional evaluation cases created by your team.
- Failures found and improvements made.

Token usage and cost report

Include evaluation/usage_report.md in code.zip. It must describe the final full-dataset run that produced output.csv, including:

- Model providers and model names.
- Number of model calls.
- Input tokens, output tokens, and total tokens.
- Average tokens per request.
- Estimated total cost and per-request cost.
- Per-model and overall totals if multiple models were used.

KEEP YOUR SUBMISSION SAFE

Do not include API keys, credentials, tokens, private data, or sensitive configuration values in prompts, code, configuration files, or the transcript.

05 / BUILD, PACKAGE & SUBMIT

Participant workflow

1. Read the repository instructions and the complete problem statement before implementing your solution.
2. Build your solution to read participant-facing files from dataset/ and write the final output.csv at the repository root.
3. Use sample_deletion_requests.csv and sample_output.csv to validate your parsing and output format. Do not use organizer-only truth or hidden answers.
4. Evaluate your approach, record the required metrics and custom test cases, and document token usage and estimated cost for the final full-dataset run.
5. Package only the runnable code, README, prompts/configuration, and evaluation reports. Exclude dataset files, virtual environments, caches, and nested archives.
6. Validate the final CSV, ZIP contents, and chat transcript before uploading the required items.

Chat transcript requirements

The repository includes AGENTS.md. Compatible AI coding tools are instructed to append conversation summaries to log.txt in the repository root. Launch the tool from the repository root (or explicitly ask it to read ./AGENTS.md first) so each turn is recorded.

Each logged turn must include the sections User Prompt, Agent Response Summary, Actions, and Context. Upload the resulting log.txt as the chat_transcript submission. Append to the log; do not rewrite earlier entries.

Final validation checklist

- ☐ output.csv header is exactly: case_id,deletion_status,steps_json.
- ☐ There are exactly 518 data rows, sorted by case_id.
- ☐ Every steps_json value parses as a JSON array; step numbers are consecutive from 1.
- ☐ Every operation/state pair is allowed and steps are ordered by phase, operation, system_id, and resource_id.
- ☐ COMPLETE cases contain verification steps only, supported by evidence.
- ☐ INCOMPLETE cases contain findings, a complete remediation plan, and future verification steps; proposed steps use evidence_id: "".
- ☐ NOT_APPLICABLE cases use [].
- ☐ code.zip contains runnable code, README.md, evaluation/evaluation_report.md, and evaluation/usage_report.md.
- ☐ The transcript follows the required turn format and contains no secrets.
- ☐ No dataset, virtual environment, cache, or nested archive is included in code.zip.

Repository & challenge reference

Official repository: https://github.com/interviewstreet/hackerrank-orchestrate-october26

Full problem statement: https://github.com/interviewstreet/hackerrank-orchestrate-october26/blob/main/problem_statement.md

Challenge page: https://www.hackerrank.com/contests/hackerrank-orchestrate-october26/challenges/ghost-data-still-exists

This participant brief is a formatted guide to the public challenge specification. In case of any discrepancy, follow the current official repository and challenge instructions.